

Muscle: Mechanisms of Contraction and Neural Control

Human Physiology - Chapter 12

Teaching Resources

December 24, 2025

Outline

12.1 Skeletal Muscles

12.2 Mechanisms of Contraction

12.3 Contractions of Skeletal Muscles

12.4 Energy Requirements of Skeletal Muscles

12.5 Neural Control of Skeletal Muscles

12.6 Cardiac and Smooth Muscles

Summary

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Summary

Structure of Skeletal Muscles

- **Skeletal muscles** (骨骼肌) produce movement through coordinated contractions
- **Origin** (起点): The stationary attachment point
- **Insertion** (止点): The movable attachment point
- **Agonists** (主动肌): Muscles causing desired movement
- **Antagonists** (拮抗肌): Muscles producing opposite movement
- **Synergists** (协同肌): Muscles assisting agonists
- **Flexors** (屈肌) decrease joint angle; **Extensors** (伸肌) increase joint angle

Structure of Skeletal Muscles

- **Skeletal Muscle Actions:** Key terminology for understanding muscle function and movement coordination



Structure of Skeletal Muscles

- Muscles are organized in a hierarchical structure with multiple layers of connective tissue
- **Epimysium** (肌外膜) surrounds the entire muscle
- **Perimysium** (肌束膜) surrounds bundles called **fascicles** (肌束)
- **Endomysium** (肌内膜) surrounds individual muscle fibers
- **Tendons** (肌腱) connect muscles to bones through continuous connective tissue

Structure of Skeletal Muscles

- The hierarchical organization of skeletal muscle from whole muscle to individual fibers
- Connective tissues provide structural support and transmit contractile force



Structure of Skeletal Muscles

- Each muscle fiber is enclosed by the **sarcolemma** (肌膜), a specialized plasma membrane
- The cytoplasm is called **sarcoplasm** (肌浆)
- Muscle fibers are **multinucleate** (多核的), formed by fusion of embryonic **myoblast cells** (成肌细胞)
- **Dystrophin** (肌营养不良蛋白) connects the sarcolemma to internal cytoskeleton
- Mutations in dystrophin cause **muscular dystrophy** (肌营养不良症)

Structure of Skeletal Muscles

- Skeletal muscle fibers display characteristic **striations** (横纹) under microscopy
- **A bands** (A 帯) appear dark; **I bands** (I 帯) appear light
- Alternating bands create the striated appearance
- Nuclei are located at the periphery of muscle fibers



Structure of Skeletal Muscles

- **Myofibrils** (肌原纤维) are the contractile units within muscle fibers
- Each muscle fiber contains hundreds to thousands of myofibrils
- Myofibrils run parallel along the length of the fiber
- They contain the proteins responsible for muscle contraction



Motor Units and Neural Control

- The **neuromuscular junction** (神经肌肉接头) is where motor neurons communicate with muscle fibers
- **Motor end plate** (运动终板): Specialized region of sarcolemma at the synapse
- **Acetylcholine (ACh)** (乙酰胆碱) is the neurotransmitter that triggers muscle contraction
- **Botulinum toxin** (肉毒杆菌毒素) blocks ACh release, causing paralysis
- Medical uses: treating muscle spasms, cosmetic applications (Botox)

Motor Units and Neural Control

- Detailed structure of the neuromuscular junction showing the synaptic cleft and motor end plate
- Synaptic vesicles contain acetylcholine ready for release



Motor Units and Neural Control

- A **motor unit** (运动单位) consists of one motor neuron and all the muscle fibers it innervates
- All fibers in a motor unit contract simultaneously when the neuron fires
- **Innervation ratio** (神经支配比): Number of muscle fibers per motor neuron
- Small motor units (low ratio) provide fine control (e.g., eye muscles: 1:23)
- Large motor units (high ratio) provide power (e.g., leg muscles: 1:1000)
- **Recruitment** (募集): Activating progressively larger motor units for stronger contractions

Motor Units and Neural Control

- Motor unit organization showing how one motor neuron branches to innervate multiple muscle fibers
- This arrangement allows graded control of muscle force through recruitment



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Summary

Sarcomere Structure

- **Myofibrils** contain **myofilaments** (肌丝): the contractile proteins
- **Thick filaments** (粗肌丝): 110 Å diameter, composed of **myosin** (肌球蛋白)
- **Thin filaments** (细肌丝): 50-60 Å diameter, composed of **actin** (肌动蛋白)
- **A bands** contain thick filaments (dark)
- **I bands** contain only thin filaments (light)
- **H bands** (H 带): Central region of A bands with only thick filaments
- **Z discs** (Z 盘): Anchor points for thin filaments

Sarcomere Structure

- Electron microscopy reveals the molecular basis of muscle striations
- The banding pattern results from the organized arrangement of thick and thin filaments



Sarcomere Structure

- A **sarcomere** (肌节) is the basic functional unit of contraction
- Extends from one Z disc to the next Z disc
- Sarcomeres are arranged in series along the myofibril
- In cross-section, thin filaments surround thick filaments in hexagonal arrays
- This arrangement allows optimal force generation

Sarcomere Structure

- Three-dimensional sarcomere architecture showing longitudinal and cross-sectional views
- The hexagonal arrangement of filaments is essential for efficient contraction



Sarcomere Structure

- **M lines** (M 线) anchor thick filaments at the sarcomere center
- **Titin** (肌联蛋白): Giant elastic protein spanning from M line to Z disc
- Titin stabilizes thick filament position within the sarcomere
- Titin provides elastic recoil, helping muscles return to resting length
- Titin is one of the largest proteins known in biology

Sarcomere Structure

- Structural proteins that maintain sarcomere organization and provide elasticity
- M lines and titin work together to stabilize the contractile machinery



Sliding Filament Theory

- The **sliding filament theory** (滑动丝理论) explains muscle contraction
- Filament lengths remain constant during contraction
- Thin filaments slide past thick filaments toward the sarcomere center
- This pulls Z discs closer together, shortening the sarcomere
- During contraction:
 - A bands: constant length
 - H bands: decrease (may disappear)
 - I bands: decrease
 - Distance between Z discs: decreases

Sliding Filament Theory

- Summary of structural changes during the sliding filament mechanism



Sliding Filament Theory

- Visual demonstration of the sliding filament model at three stages
- Progressive shortening of sarcomeres produces muscle contraction



Cross-Bridge Cycle

- **Cross-bridges** (横桥) are myosin heads that cyclically attach to and pull actin filaments
- The myosin head contains:
 - **Actin-binding site** (肌动蛋白结合位点)
 - **ATP-binding site** (ATP 结合位点) with **ATPase** (ATP 酶) activity
- The cross-bridge cycle is powered by ATP hydrolysis
- Each cycle produces a small movement; many cycles produce muscle shortening

Cross-Bridge Cycle

- **Step 1 - Activation:** ATP binds to myosin and is hydrolyzed to ADP + Pi
- This energizes the myosin head, causing it to adopt a “cocked” position
- The activated myosin head is now ready to bind to actin



Cross-Bridge Cycle

- **Step 2 - Attachment and Power Stroke:** The energized myosin head binds to actin
- Release of P_i triggers the **power stroke** (动力冲程)
- The myosin head changes angle, pulling the actin filament
- ADP is then released, completing the power stroke



Cross-Bridge Cycle

- **Step 3 - Detachment:** A new ATP molecule binds to the myosin head
- This causes the myosin to detach from actin
- The cycle repeats as long as ATP and Ca^{2+} are available
- Without ATP, myosin remains bound to actin (**rigor mortis** 尸僵)



Regulation of Contraction

- Muscle contraction is regulated by calcium ions and regulatory proteins
- **Tropomyosin** (原肌球蛋白): Rod-shaped protein that lies along actin strands
- **Troponin** (肌钙蛋白): Three-subunit complex attached to tropomyosin
 - **Troponin T**: Binds to tropomyosin
 - **Troponin I**: Inhibits actin-myosin interaction
 - **Troponin C**: Binds Ca^{2+}
- At rest, tropomyosin blocks myosin-binding sites on actin

Regulation of Contraction

- Structural organization of the thin filament regulatory system
- Troponin and tropomyosin work together to control access to actin-binding sites



Regulation of Contraction

- When Ca^{2+} is present, it binds to troponin C
- This causes the troponin-tropomyosin complex to shift position
- Myosin-binding sites on actin are exposed
- Cross-bridge cycling can now occur, producing contraction
- Removal of Ca^{2+} causes relaxation

Regulation of Contraction

- Calcium-induced conformational changes expose actin-binding sites
- This calcium-troponin interaction is the key regulatory step in muscle contraction



Sarcoplasmic Reticulum and Excitation-Contraction Coupling

- The **sarcoplasmic reticulum (SR)** (肌浆网) stores and releases Ca^{2+}
- Modified endoplasmic reticulum specialized for calcium regulation
- **Terminal cisternae** (终池): Enlarged SR regions with high Ca^{2+} concentration
- **T-tubules** (横小管): Invaginations of the sarcolemma penetrating into the fiber
- **Triad** (三联体): One T-tubule flanked by two terminal cisternae
- T-tubules allow rapid transmission of action potentials to the fiber interior

Sarcoplasmic Reticulum and Excitation-Contraction Coupling

- The relationship between sarcoplasmic reticulum, T-tubules, and myofibrils
- This arrangement enables rapid, synchronized calcium release throughout the muscle fiber



Sarcoplasmic Reticulum and Excitation-Contraction Coupling

- **Excitation-contraction coupling** (兴奋-收缩耦联) links electrical excitation to mechanical contraction
- Key steps:
 1. ACh triggers action potential on sarcolemma
 2. Action potential propagates along sarcolemma and down T-tubules
 3. Voltage-gated channels in T-tubules activate
 4. This triggers Ca^{2+} release from SR
 5. Ca^{2+} binds to troponin, initiating contraction
 6. Ca^{2+} pumps return calcium to SR, causing relaxation

Sarcoplasmic Reticulum and Excitation-Contraction Coupling

- Detailed mechanism of excitation-contraction coupling in skeletal muscle
- The process involves mechanical coupling between T-tubule and SR proteins



Sarcoplasmic Reticulum and Excitation-Contraction Coupling

- Summary diagram showing how electrical signals couple to calcium release
- This elegant mechanism ensures rapid, coordinated muscle contraction



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Summary

Twitch, Summation, and Tetanus

- A **twitch** (单收缩) is the response to a single action potential
- Three phases:
 - **Latent period** (潜伏期): 2 ms delay before contraction
 - **Contraction phase** (收缩期): Tension development
 - **Relaxation phase** (舒张期): Return to resting state
- Single twitches rarely occur naturally in the body

Twitch, Summation, and Tetanus

- Recording techniques demonstrate the mechanical response to electrical stimulation
- Twitches are the building blocks of more complex contractions



Twitch, Summation, and Tetanus

- **Summation** (总和): Repeated stimulation before complete relaxation
- **Incomplete tetanus** (不完全强直收缩): Moderate frequency (5-10/sec)
 - Individual twitches visible but summate
- **Complete tetanus** (完全强直收缩): High frequency (60+/sec)
 - Smooth, sustained contraction
 - No visible individual twitches
- **Muscle fatigue** (肌肉疲劳) eventually occurs with prolonged stimulation

Twitch, Summation, and Tetanus

- Progressive increase in muscle tension with increasing stimulation frequency
- Complete tetanus produces maximum sustainable force



Types of Muscle Contractions and Mechanical Properties

- **Isotonic contraction** (等张收缩): Muscle length changes, tension constant
 - **Concentric** (向心): Muscle shortens (lifting)
 - **Eccentric** (离心): Muscle lengthens while contracting (lowering)
- **Isometric contraction** (等长收缩): Tension develops, length constant
 - Occurs when load exceeds muscle's ability to shorten
 - Important for maintaining posture
- **Series-elastic component** (串联弹性成分): Tendons, titin, cross-bridges

Types of Muscle Contractions and Mechanical Properties

- The **force-velocity relationship** (力-速度关系) shows inverse relationship
- Greater load → slower shortening velocity
- Lighter load → faster shortening velocity
- Maximum velocity occurs with minimal load
- Isometric contraction occurs when load prevents shortening



Types of Muscle Contractions and Mechanical Properties

- The **length-tension relationship** (长度-张力关系) depends on sarcomere length
- **Optimal length** (最适长度): 2.0-2.25 μm sarcomere length
 - Maximum filament overlap
 - Maximum cross-bridge formation
- Shorter than optimal: Filament interference, compression against Z discs
- Longer than optimal: Reduced overlap, fewer cross-bridges

Types of Muscle Contractions and Mechanical Properties

- Maximum tension occurs at optimal sarcomere length with ideal filament overlap
- This relationship explains why muscle strength varies with joint angle



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Metabolism of Skeletal Muscles

- Muscle contraction requires ATP for:
 - Cross-bridge cycling (myosin ATPase)
 - Ca^{2+} pumping (SERCA pumps)
 - $\text{Na}^+ - \text{K}^+$ pumping (maintaining membrane potential)
- Multiple energy sources are used depending on exercise intensity and duration

Metabolism of Skeletal Muscles

- **Phosphocreatine (PCr)** (磷酸肌酸) provides immediate energy reserve
- Reaction: $\text{PCr} + \text{ADP} \rightleftharpoons \text{Cr} + \text{ATP}$
- Catalyzed by **creatine phosphokinase (CPK)** (肌酸磷酸激酶)
- Rapidly replenishes ATP during first 5-10 seconds of intense activity
- Allows ATP levels to remain nearly constant initially



Metabolism of Skeletal Muscles

- Energy source usage varies with exercise intensity
- **Plasma glucose** (血糖): Increases during exercise
- **Muscle glycogen** (肌糖原): Major fuel for moderate-intense exercise
- **Plasma free fatty acids** (血浆游离脂肪酸): Important for prolonged mild exercise
- **Muscle triglycerides** (肌肉三酰甘油): Used during moderate exercise

Metabolism of Skeletal Muscles

- Relative contributions of different fuel sources change with exercise intensity
- Understanding fuel utilization helps optimize training and performance



Metabolism of Skeletal Muscles

- Muscle contraction increases glucose uptake independent of insulin
- **GLUT4 transporters** (GLUT4 转运蛋白) translocate to sarcolemma during contraction
- Uptake increases with both exercise intensity and duration
- This mechanism is important for blood glucose regulation



Fiber Types and Adaptations

- Skeletal muscles contain different fiber types with distinct properties
- **Type I (Slow Oxidative)** (I 型/慢氧化型):
 - Slow contraction, high endurance
 - High mitochondria, myoglobin, capillaries
 - Red color, fatigue-resistant
 - Used for posture, endurance activities
- **Type IIA (Fast Oxidative-Glycolytic)** (IIA 型):
 - Fast contraction, moderate endurance
 - Intermediate properties
 - Can use aerobic and anaerobic metabolism

Fiber Types and Adaptations

- **Type IIX (Fast Glycolytic)** (IIX 型/快糖酵解型):
 - Very fast contraction, low endurance
 - Low mitochondria, high glycogen
 - White color, fatigue rapidly
 - Used for brief, intense activities
- Fiber type distribution varies among individuals due to genetics and training

Fiber Types and Adaptations

- Detailed comparison of fiber type characteristics
- Different fiber types are specialized for different functional demands



Fiber Types and Adaptations

- Contraction speed varies dramatically among muscle fiber types
- Fast-twitch fibers develop maximum tension much more rapidly than slow-twitch



Fiber Types and Adaptations

- Fiber type composition varies tremendously among individuals
- Influenced by both genetics and physical training
- Elite athletes often show extreme fiber type distributions suited to their sport



Fiber Types and Adaptations

- **Endurance training** (耐力训练) produces specific adaptations:
 - Increased mitochondrial density and enzymes
 - Increased capillary density
 - Increased myoglobin content
 - Enhanced fat oxidation capacity
 - Increased $\dot{V}O_{2\max}$ (最大摄氧量)
 - Shift toward Type I and Type IIA fibers

Fiber Types and Adaptations

- Comprehensive metabolic and structural changes with endurance training
- These adaptations improve aerobic capacity and fatigue resistance



Fiber Types and Adaptations

- **Resistance training** (抗阻训练) produces different adaptations:
 - Muscle fiber **hypertrophy** (肌肥大)
 - Increased myofibril number
 - Increased contractile protein synthesis
 - Increased strength and power
 - Minimal changes in aerobic capacity
- **Satellite cells** (卫星细胞) are muscle stem cells enabling repair and growth

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Neural Control Hierarchy

- Skeletal muscle control involves multiple levels of neural organization
- **Lower motor neurons** (下运动神经元): Directly innervate muscle fibers
 - Alpha motor neurons: Innervate extrafusal (regular) muscle fibers
 - Gamma motor neurons: Innervate intrafusal (spindle) fibers
- **Upper motor neurons** (上运动神经元): Control lower motor neurons from brain/brainstem

Neural Control Hierarchy

- Terminology for understanding the neural control hierarchy
- This organization allows both reflex and voluntary control of movement



Muscle Spindles and Proprioception

- **Muscle spindles** (肌梭) are stretch receptors detecting muscle length changes
- Contain specialized **intrafusal fibers** (梭内纤维) in parallel with regular **extrafusal fibers** (梭外纤维)
- Two types of intrafusal fibers:
 - **Nuclear bag fibers** (核袋纤维)
 - **Nuclear chain fibers** (核链纤维)
- Sensory innervation:
 - **Type Ia afferents** (Ia 类传入神经): Detect rate and magnitude of stretch
 - **Type II afferents** (II 类传入神经): Detect static stretch

Muscle Spindles and Proprioception

- Detailed anatomy of muscle spindles and their innervation
- Spindles provide continuous information about muscle length to the CNS



Muscle Spindles and Proprioception

- **Alpha motor neurons** (α 运动神经元):
 - Large diameter, fast conduction
 - Innervate extrafusal fibers
 - Cause muscle contraction
- **Gamma motor neurons** (γ 运动神经元):
 - Smaller diameter, slower conduction
 - Innervate intrafusal fibers
 - Adjust spindle sensitivity
- **Alpha-gamma coactivation** (α - γ 协同激活): Both activated simultaneously
 - Maintains spindle tension during muscle contraction
 - Ensures continuous length monitoring

Muscle Reflexes

- The **stretch reflex** (牵张反射) is a **monosynaptic** (单突触) reflex
- Resists unwanted muscle lengthening
- Sequence:
 1. Muscle stretched
 2. Spindles activated
 3. Type Ia afferents fire
 4. Direct synapse on alpha motor neurons
 5. Muscle contracts
- Fastest reflex in the body

Muscle Reflexes

- Summary of stretch reflex events
- This reflex provides rapid, automatic resistance to muscle stretch



Muscle Reflexes

- The **knee-jerk reflex** (膝跳反射) is a classic stretch reflex example
- Tapping the patellar tendon stretches the quadriceps
- Muscle spindles activate, causing immediate contraction
- Used clinically to assess nervous system function



Muscle Reflexes

- **Golgi tendon organs (GTOs)** (高尔基腱器官) detect muscle tension
- Located in tendons, sensitive to force rather than length
- When tension is excessive:
 - GTOs activate Type Ib afferents
 - Inhibitory interneurons activated (**disynaptic** 双突触)
 - Motor neuron activity reduced
 - Muscle relaxes (protective mechanism)
- This is the **Golgi tendon reflex** (高尔基腱反射)

Muscle Reflexes

- GTO-mediated inhibition protects against excessive muscle force
- This negative feedback prevents muscle and tendon damage



Muscle Reflexes

- **Reciprocal innervation** (交互神经支配) coordinates agonist-antagonist pairs
- Spindle afferents from agonist muscle:
 - Excite alpha motor neurons to agonist (direct)
 - Inhibit alpha motor neurons to antagonist (via inhibitory interneuron)
- Ensures smooth, coordinated movement
- Prevents opposition between agonist and antagonist

Muscle Reflexes

- Neural circuitry for reciprocal innervation
- This wiring pattern is fundamental to coordinated motor control



Muscle Reflexes

- The **crossed-extensor reflex** (交叉伸肌反射) coordinates bilateral limb movements
- Painful stimulus causes:
 - Ipsilateral (same side) withdrawal (flexion)
 - Contralateral (opposite side) extension
- Demonstrates **double reciprocal innervation** (双重交互神经支配)
- Polysynaptic, complex reflex for coordinated response

Muscle Reflexes

- Complex reflex coordination for protective withdrawal response
- Involves multiple spinal cord segments and interneurons



Muscle Reflexes

- Upper motor neurons originate in motor cortex and brainstem
- Descending pathways:
 - **Pyramidal tracts** (锥体束): Direct voluntary control
 - **Extrapyramidal tracts** (锥体外系): Posture and automatic movements
- **Upper motor neuron damage** (上运动神经元损伤):
 - Spastic paralysis, hyperactive reflexes, Babinski sign
- **Lower motor neuron damage** (下运动神经元损伤):
 - Flaccid paralysis, absent reflexes, muscle atrophy

Muscle Reflexes

- Clinical signs distinguishing upper from lower motor neuron lesions
- Understanding these patterns aids in neurological diagnosis



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Summary

Cardiac Muscle

- **Cardiac muscle** (心肌) shares features with skeletal muscle:
 - Striated appearance
 - Sarcomere organization
 - Troponin-tropomyosin regulation
- Unique features:
 - Short, branched cells
 - Uninucleate or binucleate
 - **Intercalated discs** (闰盘) connecting cells
 - Forms **functional syncytium** (功能性合胞体)

Cardiac Muscle

- **Gap junctions** (缝隙连接) allow electrical coupling between cardiac cells
- Composed of **connexin proteins** (连接蛋白)
- Enable coordinated contraction of the heart
- Located primarily at intercalated discs



Cardiac Muscle

- Cardiac muscle structure showing intercalated discs
- The branched architecture and cell-cell connections create an integrated network



Cardiac Muscle

- Cardiac muscle excitation-contraction coupling differs from skeletal muscle
- Requires extracellular Ca^{2+} in addition to SR calcium
- **Ca^{2+} -induced Ca^{2+} release** (钙诱导钙释放):
 - Small amount of Ca^{2+} enters from outside
 - This triggers larger release from SR
 - Amplifies the calcium signal
- Allows graded contractions based on extracellular calcium

Cardiac Muscle

- Detailed mechanism of cardiac excitation-contraction coupling
- The two-source calcium system allows for more flexible regulation



Smooth Muscle

- **Smooth muscle** (平滑肌) has distinct structural features:
 - Non-striated appearance
 - Spindle-shaped cells
 - Single central nucleus
 - No organized sarcomeres
- Thick and thin filaments attach to **dense bodies** (致密体)
- Dense bodies are analogous to Z discs
- **Intermediate filaments** (中间丝) form cytoskeleton

Smooth Muscle

- Smooth muscle lacks the organized sarcomere structure
- Dense body arrangement allows contraction in multiple directions



Smooth Muscle

- Smooth muscle regulation uses **calmodulin** (钙调蛋白), NOT troponin
- Mechanism:
 1. Ca^{2+} binds to calmodulin
 2. Ca^{2+} -calmodulin activates **MLCK** (myosin light-chain kinase) (肌球蛋白轻链激酶)
 3. MLCK phosphorylates myosin light chains
 4. Phosphorylated myosin can bind actin
 5. **Myosin phosphatase** (肌球蛋白磷酸酶) ends contraction
- Allows slower, sustained contractions
- Can maintain **latch state** (锁定状态) with low energy use

Smooth Muscle

- Phosphorylation-based regulation of smooth muscle contraction
- This mechanism enables prolonged contractions with minimal energy expenditure



Smooth Muscle

- Two types of smooth muscle organization:
- **Single-unit smooth muscle** (单单位平滑肌):
 - Cells coupled by gap junctions
 - Function as syncytium
 - Spontaneous activity (pacemaker cells)
 - Found in: GI tract, uterus, small blood vessels
- **Multiunit smooth muscle** (多单位平滑肌):
 - Cells NOT coupled
 - Individual innervation required
 - Precise control
 - Found in: iris, ciliary muscle, airways

Smooth Muscle

- Contrasting organization of single-unit and multiunit smooth muscle
- Different control mechanisms suit different physiological needs



Smooth Muscle

- Comprehensive comparison of all three muscle types
- Understanding differences helps explain their specialized functions



Summary

Summary

- Skeletal muscle structure is hierarchically organized for efficient force generation
- The sliding filament mechanism and cross-bridge cycle explain contraction at molecular level
- Calcium regulation through troponin-tropomyosin controls contraction
- Excitation-contraction coupling links electrical signals to mechanical responses
- Fiber types and metabolism adapt to different functional demands
- Neural control involves reflexes, proprioception, and voluntary commands
- Cardiac and smooth muscles have specialized structures and regulation for their roles